

SONAL SHREYA

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PROFESSIONAL SUMMARY

- Software Engineer Lead with 8+ years building high-availability enterprise applications across manufacturing, financial services, healthcare, and global logistics domains using Core Java, Spring Boot, microservices and secure protocols.
- Delivered distributed systems processing millions of transactions with 99.9% uptime via REST APIs, Kafka, and multithreading. Improved performance through JPA/Hibernate optimization and automated CI/CD Docker deployments on AWS (40% faster releases).
- Led cross-functional teams delivering microservices and cloud migrations achieving 20 -40% performance gains and 40% cost reduction, also resolved production incidents, mentored developers.
- M.S. in AI & Business Analytics and Business Intelligence with focus on AI/LLMs, agentic systems, and AI-powered UX.

EDUCATION

University of South Florida, Tampa FL

Master of Science, Artificial Intelligence and Business Analytics

Graduated: **May, 2026**

Kalinga Institute of Industrial Technology University, Bhubaneswar, India

Bachelor of Technology, Computer Science and Engineering

Graduated: **Jul, 2016**

TECHNICAL SKILLS

Programming Languages: Java, Python, C#, SQL, PL/SQL, C, Unix

Backend: Microservices, APIs, Spring Boot, Spring Batch, Hibernate, Node.js, Agile, Event-Driven Architecture Spring MVC, Spring JPA, J2EE, JVM

Cloud: AWS (Lambda, ECS,EC2, S3, CloudWatch, Aurora, RDS,SNS, Kinesis), Azure (Blob, Functions,DynamoDB)

Tools & DevOps: Docker, CI/CD, Jenkins, TeamCity, GitHub, Maven, JUnit, Mockito, SonarQube, Swagger, Postman, JIRA, Confluence, Figma, SVN, Bitbucket, Grafana,

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Angular, Material UI, Chart.js

Databases and distributed: Oracle, SQL Server, PostgreSQL, MongoDB, Cassandra, Hadoop, HDFS, Snowflake, Hive

AI/ML: Pytorch, NLP, LLM, RAG, Agentic AI

Certifications: AWS Developer Associate, AZ-900, Google Product Management, Microsoft SDLC

PROFESSIONAL EXPERIENCE

University of South Florida- Tampa, Florida [Graduate Research Assistant]

Aug, 2025 – May, 2026

- Facilitated research on a GenAI RAG pipeline for healthcare text translation, reducing hallucinations.
- Worked on Agentic AI workflows and Text Analytics

Ground Game.Health - Tampa, Florida [Software Development Intern]

Jun, 2025 – Aug, 2025

- Stabilized .NET/MySQL backend (–15% bugs), built AI-ready APIs/pipelines automating 70% workflows for email and SMS communications, improved ML precision 15%, reduced processing time 40%, and delivered 4+ Power BI dashboards.

Capgemini - Noida, India [Software Engineer Lead]

May, 2022 – Jul, 2024

- Built Spring Boot microservices supporting automated manufacturing workflows for Stellantis, improving throughput 20% and reducing latency 40%.
- Developed 25+ REST APIs for web services and integrations using Swagger/OpenAPI.
- Implemented data processing logic using Apache Spark (Spark SQL, DataFrames) within backend workflows to transform and analyze large datasets.
- Led Agile Scrum ceremonies including daily standups, sprint planning, retrospectives, and cross-functional coordination to support timely delivery of microservice enhancements and releases.
- Optimized JPA/Hibernate ORM and SQL queries and PL/SQL stored procedures using ACID properties, improving 35% DB response time.
- Enforced unit testing (JUnit, Mockito) and SonarQube quality gates, ensuring 90% test coverage.
- Containerized services with Docker and automated CI/CD (Jenkins/TeamCity, Docker, Kubernetes), cutting release effort 40%; resolved critical incidents and mentored junior developers.

Accenture - Noida, India [Application Development Senior Analyst]

May, 2021 – May, 2022

- Created event-driven microservices (Java, Spring Boot, Kafka) for real-time data ingestion, improving processing efficiency 20%, troubleshooting using Unix/Linux shell commands and reducing latency via asynchronous architecture.
- Designed maintainable architectures (object oriented programming, data structures, Factory/Strategy/Singleton), enabled A/B testing for feature validation, performed code reviews and maintained 99% uptime with 90%+ Sonar quality.
- Developed backend integrations with HDFS and Hive to support large-scale data access and querying.
- Implemented version control and configuration management processes, reducing deployment errors by 5%.
- Built secure REST APIs supporting financial transaction and payment-processing workflows within enterprise banking systems, with exposure to blockchain concepts and distributed ledger-based transaction processing.
- Awarded Best People Award (FY 2021–22) for performance excellence and peer mentoring.

Infosys Limited - Bhubaneswar, India [Technology Analyst]

Jul, 2016 – May, 2021

- Delivered Java-based backend services for global liner and logistics clients (APL, CMA-CGM), delivering high-availability enterprise systems with 95% on-time milestone execution.
- Translated product requirements into scalable service architectures and REST API specifications, aligning backend design with business workflows, sprint panning and user story creations.
- Defined Application designs (LLD/HLD) for multi-tier distributed applications integrating UI, services, and database layers, Authentication & Authorization, API security, session management, cutting revision cycles 35%.
- Conducted production support, monitored logs, investigated failures, and resolved P1/P2 incidents within SLA timelines with secure coding practices and in-depth analysis, improving system efficiency 20% and strengthening reliability.

PROJECT EXPERIENCE

Urbana Threads

- Architected and deployed a full-stack retail management system (ASP.NET MVC, C#, Azure SQL) with real-time inventory tracking, sales dashboards (Chart.js), and a Botpress AI assistant on Azure App Service.

Tampa Bay County Insights Dashboard

- Developed and implemented a full-stack county insights platform (React, Python), implementing REST APIs and data pipelines for socioeconomic analysis, reducing manual reporting effort by 30%.

Stellantis AWS Migration of Applications

- Spearheaded migration of 15+ on-prem applications to AWS (EC2, S3, Lambda), achieving 99.5% success through risk assessment, code optimization, and cross-functional collaboration.

Clinical Review Assistant RAG + Agentic LLM System (Healthcare)

- Architected a scalable RAG-based backend service (Pinecone, LangGraph) for secure document retrieval and summarization, implementing distributed service architecture and citation-backed responses.

Florida Labor Market Wage Analysis

- Analyzed 10 years of wage data across 25 Florida metros using Apache Spark and PySpark.
- Created a Random Forest model with 85% accuracy to predict wages by occupation, city, and employment.
- Developed a classification model with Spark-based oversampling to resolve class imbalance.
- Conducted wage disparity analysis using SQL, SOC codes, and visual tools (choropleth maps, wage bands).